

Generation of CHARMM Topology and Parameter Files for Small Molecules Using MolAICal and MATCH Algorithms

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1. Introduction

Methods for rapidly constructing topology and parameter files of small-molecule CHARMM force fields—such as the CGENFF program—generally rely on pattern-matching algorithms. Even when using the Force Field Toolkit (FFTK) [1], atom typing still requires either automated matching or manual inspection and assignment. MolAICal integrates an upgraded and modified MATCH [2] to automate the generation of CHARMM-compatible topology and parameter files for small molecules. The core algorithm of MATCH (under MIT license) is based on graph theory and breadth-first search, parsing molecules into directed graphs and expanding each atom's local environment up to 10 bonds to capture chemical context. Automated atom typing is achieved through maximum fragment matching against a library of predefined "super SMILES" strings. Partial charges are derived via iterative deconvolution using bond charge increment (BCI) rules, while missing parameters are intelligently generated through a substitution scheme based on chemical similarity scoring. Validated through cross-validation and leave-one-out testing, the method achieved charge fitting R^2 values of 0.94–1.0 (average unsigned errors of 0.0013–0.024 e), solvation energy predictions with $R^2 = 0.99$, and excellent reproducibility for bond angles and dihedral parameters, thoroughly confirming its reliability and accuracy. This approach is well-suited for high-throughput virtual screening, extension and gap-filling of existing force fields, and cross-force-field validation, supporting input formats including PDB, MOL, SDF, and MOL2. It enables rapid batch generation of CHARMM-compatible files to meet large-scale preliminary screening demands; however, for studies requiring higher precision—such as rigorous free energy calculations or production-grade molecular dynamics simulations—it is recommended to subsequently refine parameters using the Force Field Toolkit (FFTK) or FFParam [3] for QM-level optimization incorporating quantum mechanical calculations and experimental data.

2. Materials

2.1. Software requirement

- 1) MolAICal: <https://molaical.github.io>
- 2) UCSF Chimera: <https://www.cgl.ucsf.edu/chimera>

2.2. Example files

- 1) All the necessary tutorial files are downloaded from:
<https://github.com/MolAICal/tutorials/tree/master/032-gencharmmff>

3. Procedure

3.1. Generate the CHARMM topology and parameter files for small molecules

Go to the folder [032-gencharmmff](#), and run the command below:

- 1) Ensure the atom names in the file are unique. If atom names are the same, it will be impossible

to know which atoms are linked, causing the CHARMM topology file to be distorted or erroneous.

Use the following command:

```
#> molaical.exe -call run -c sfile -i u_atom_name.py ligand.pdb ligand.pdb
```

- ◆ **1st:** the first parameter is the input molecular file name;
- ◆ **2nd:** the second parameter is the output molecular file name after changing the atoms to unique names, with coordinates remaining unchanged.

2) Then generate the CHARMM force field for molecule:

```
#> molaical.exe -call charmm -i ligand.pdb
```

The output file after running is a file with the .str suffix containing the CHARMM topology and parameters of the input molecules, which can be subsequently used to construct .pdb and .psf files for NAMD molecular dynamics (MD) simulations.

Option: If researchers want to separately generate a topology file (suffix .rtf) and a parameter file (suffix .prm) instead of a merged file with the suffix .str that combines topology and parameters, they can add the parameter "-div" and run the command as follows:

```
#> molaical.exe -call charmm -i ligand.mol2 -div
```

Note: the input molecule can also be "ligand.pdb". Here, it just shows that the input molecule can be in the PDB or MOL2 format.

Deep study:

If run:

```
# Make the atom names in the molecule unique
```

```
#> molaical.exe -call run -c sfile -i u_atom_name.py final_lig_138.mol2 final_lig_138.mol2
```

```
# Generate the topology and parameter file
```

```
#> molaical.exe -call charmm -i final_lig_138.mol2
```

The molecule **final_lig_138.mol2** has been optimized within the active pocket **rather than undergoing energy minimization in the free state**. As a result, some interatomic distances have changed, **which may cause the matching algorithm to fail in assigning charges or other parameters**, potentially leading to the following error:

Could not determine a suitable solution for bond types for UNL1! be wary

The Following Bonds Could Not be Charged!

Name	Type	Name	Type	Bond Type
C	CG2R5I	C	CG3II	1
C	CG2DI	N	NG3II	1

C CG2D1 N NG311 I
Could not Finish Charging of Molecule! at E:/test/log/MolAICal-
win64//mtools/matchvx/MATCH/lib/MATCHer.pm line 174.

OR

Allowed Protonation Not defined for Cl for C in <0> at E:/test/log/MolAICal-
win64//mtools/matchvx/PerlChemistry/lib/Molecule.pm line 515.

Solution:

Users can use UCSF Chimera to perform energy minimization on a single molecule, then save it as a PDB file named 138.pdb (as shown in the Figure 1 below), and then run the above command again

```
#> molaical.exe -call charmm -i 138.pdb
```

to resolve this problem.

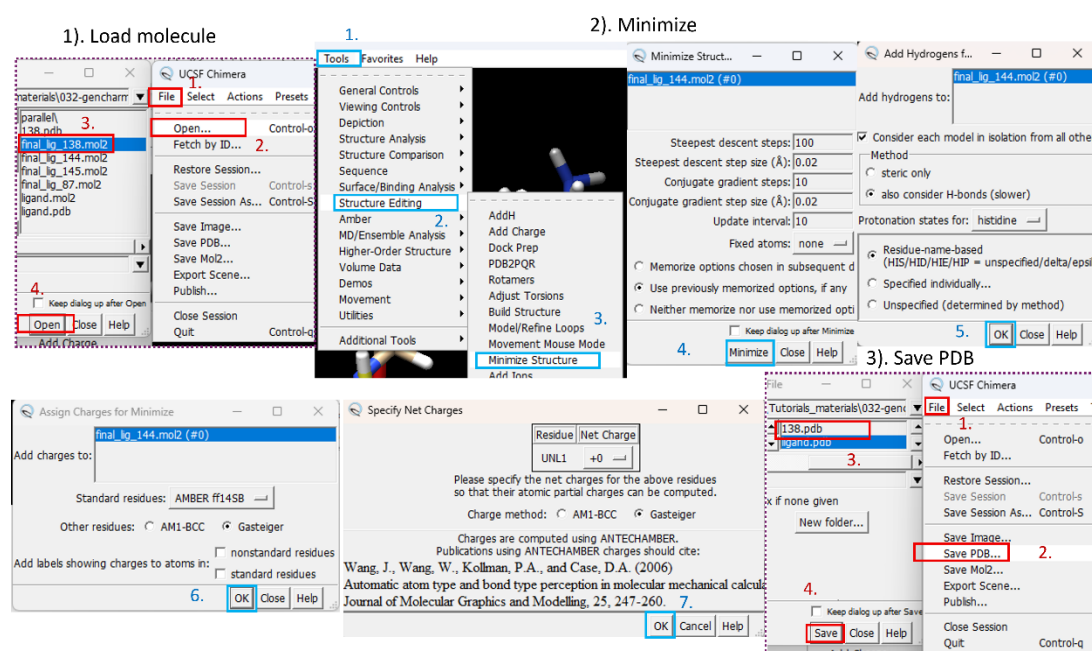


Figure 1. Minimize molecule

3.2. Generate the CHARMM topology and parameter files for small molecules with CPU multi-cores

Go to the folder `032-gencharmff/parallel`, and run the command below:

```
# Make the atom names in the molecule unique in parallel mode
```

```
#> molaical.exe -call run -c parallel -nc 3 -n "*.pdb" -i molaical.exe -call run -c sfile -i u_atom_name.py marg marg
```

```
# Generate the topology and paramter file in the parallel mode
```

```
#> molaical.exe -call run -c parallel -nc 3 -n "*.pdb" -i molaical.exe -call charmm -i marg
```

Note:

- ◆ **-call:** Interact with external programs or commands. '**run**' means to call the external programs, which can search the path of the program from the set environment file.
- ◆ **-i :** The command arguments of the program. It includes **all command-line arguments of external programs**, but the '-i' parameter must be the last one specified, while all other identifiers (e.g., '-p', etc.) of MolAICal must come **before '-i'**.
- ◆ **-n:** The specified keyword (here is “*.pdb”) is used to find filenames containing it in a designated directory; those filenames then substitute the 'marg' placeholder in an external command, enabling batch and parallel file processing. Default match rule is '**SMART**'.
- ◆ **-nc <arg>** The number of calling CPU cores.
- ◆ **-c:** It will invoke the external command lines in the parallel running mode if its value is "**parallel**".

The run process: first find all molecular filenames containing the ".pdb" string in the current directory (including all subdirectories); then substitute these filenames for the "marg" placeholder — one filename per command line, replacing all occurrences of "marg"; after that, perform parallel computation using multiple CPU cores (in this case, 3 CPU cores with -nc 3). Based on this rule, users can easily extend the approach to running other external command-line programs in parallel.

References

1. Mayne CG, Saam J, Schulten K, Tajkhorshid E, Gumbart JC. Rapid Parameterization of Small Molecules Using the Force Field Toolkit. *Journal of Computational Chemistry*. 2013;34(32):2757 - 70. doi: 10.1002/jcc.23422. PubMed PMID: WOS:000326466900001.
2. Yesselman JD, Price DJ, Knight JL, Brooks CL. MATCH: An Atom-Typing Toolset for Molecular Mechanics Force Fields. *J Comput Chem*. 2012;33(2):189-202. doi: 10.1002/jcc.21963. PubMed PMID: WOS:000297852600008.
3. Kumar A, Yoluk O, MacKerell A. FFParm: Standalone package for CHARMM additive and Drude polarizable force field parametrization of small molecules. *Journal of Computational Chemistry*. 2020;41(9):958-70. doi: 10.1002/jcc.26138. PubMed PMID: WOS:000504769500001.